Introduction

Mobile Newsmaking

> "News is a window on the world." (Tuchman 1978, p. 1)

We expect our morning newspaper, the hourly news radio broadcasts, and the evening TV news programs, as well as continuously updated news websites, to provide us with such a window on the world everyday. However, we seldom reflect upon the work behind it.

In the sixties and seventies, sociologists like Herbert Gans (1979) and Gaye Tuchman (1978) wandered into American newsrooms to ask a simple question: How is news made? Their ethnographic studies offer detailed descriptions of the work behind "the window on the world". They also highlight the complex relationship between the organisation of news production and the technology in use.

In the nineties, Internet and World Wide Web gave us new types of windows on the world. Not only has the news medium changed, Internet has also improved the technology used in the newsroom, changed the organisation of news production, and provided new ways to send and receive news material for reporters in the field. Many argue that this is just the beginning:

> "Ultimately, despite the attention being given to how the Internet is influencing the future of newsrooms and news coverage, a much greater impact on how reporters do their jobs will likely result from continued advances in electronics miniaturization and the development of easier to use and innovative multiple-media news-gathering tools."

This was written by Kerry J Northrup (2000a) in a series of reports on NewsGear, an initiative of the IFRA Center for Advanced News Operations. NewsGear compress the technologies a mobile reporter would need into a regulation-size airline travel case (see figure 1).



Figure 1: NewsGear, 2001 “special edition” (Cole, 2001)

In this thesis I argue that easy to use mobile devices with innovative IT services will have great impact not only on how reporters gather news items, send and retrieve news material, but also on how reporters *make news stories*. That is, how they transform everyday occurrences and happenings into newsworthy stories, discuss news events and formulate news tasks, do background research and frame news stories.

For many journalists it is a way of life to file news stories from hotel rooms, airports, catastrophe spots and chaotic press conferences far from their newsrooms. Reporters get the job done at just about any time there is a pause in the schedule and in any space large enough to set up a portable computer. There is also a type of mobile reporting that does not have the same spectacular characteristics as that of journalists moving from one big event to another big event. Many reporters never leave their neighbourhood, but still work much of their time out of the newsroom.

In the late nineties, we wandered into Swedish newsrooms and followed news reporters when they left the newsroom to cover assignments in their hometown or in the nearest region (cf., Fagrell and Ljungberg, 2000). We asked two questions: How is news made? How to design IT support for mobile news journalism?

Not only mobile news reporters, but also all of us who are part of the information society are exploiting emerging information technologies such as Internet. The rapid diffusion of mobile communication technologies such as cell phones and personal digital

assistants (PDAs) have made it possible for many people to become 'digital nomads' (Makimoto and Manners, 1997). However, these technologies also takes the general problems of designing user interface from stationary settings out to more demanding environments (cf., Kristoffersen and Ljungberg, 1999). The lack of time, silence and focus in mobile situations means that mobile IT services must provide users with *timely knowledge for the task at hand* (Fagrell, 2000). In my research I have focused upon one specific aspect of this, i.e. how to provide mobile reporters with *contextual information relevant for the task at hand*.

Contextual information reminds the user of the context for the task at hand, helps the user to understand, make decisions and act. For example, a mobile reporter should be provided with an overview of internal and external information related to the current news tasks and get suggestions of knowledgeable and available colleagues to contact. Such a PDA-based (see figure 2) application keeps reporters updated also when they leave the newsroom. It will help the reporters to ask relevant and interesting questions and to properly frame the news stories. This kind of design concept and IT service exemplify the results from the research reported in this thesis.



Figure 2: FieldWise (see *Paper 3* and *4*) in use

1. The thesis

The research reported in this thesis has been part of the CSCW (*Computer Supported Cooperative Work*) research area. The research approach that has been used is *Informatics*. Informatics, as it has been applied in the Mobile Informatics programme at the Viktoria institute, is a design-oriented study of information technology use.

The first paper deals with the overall task of *designing new and innovative IT services for mobile news journalism*. The next three papers address the specific research question: *How to provide reporters with contextual information for news tasks in mobile situations?*

In the following, I will introduce these papers by first describing the research background using four keywords: *news* and *newsmaking*, *mobility* and *context*. Firstly, I will depict the general concept of "news" and the domain of work studied, i.e. work conducted by journalists to make news. Secondly, I will give a background to the issue of "mobility", especially the contextual dimension of mobile work and interaction. In addition, the introduction offers an overview of the research process, an outline of the papers, a summary of the contributions and how they respond to the research question. Finally, the first chapter concludes with a discussion on contractions revealed throughout the research in the borderline between academia and industry.

The four individual papers of the thesis are:

1. Forsberg, K. (1999) "Navigating in NewsSpace," In *Proceedings of Computer Supported Cooperative Work on Design*, CSCWD99, pp. 329-336, Compiègne, France.

2. Fagrell, H., K. Forsberg, E. Johannesson and F. Ljungberg (2000) "NewsMate: Providing Timely Information to Mobile and Distributed News Journalists," In *Extended Abstract of the ACM 2000 Conference on Human Factors in Computing Systems*, pp. 121-122, The Hauge, The Netherlands: ACM Press.

3. Fagrell, H., K. Forsberg and J. Sanneblad (2000) "FieldWise: a Mobile Knowledge Management Architecture," In *Proceedings of ACM 2000 Conference on Computer Supported Cooperative Work*, pp. 211-220, Philadelphia, PA: ACM Press.

4. Forsberg, K. (2001) "Localising News: Task Articulation and Contextualisation in Mobile News Journalism,"[1] Submitted to *Journal of Graphic Technology*.

Other publications of mine that are related to the thesis:

1. Börjesson, M., I. Bruce, K. Forsberg, (2000) "Academia, Industry and Consulting firms in Collaboration, An value-based analysis of different institutions," In *Proceedings of the 23rd Information Systems Research in Scandinavia (IRIS 23)*, Vol. 1, pp. 333-340, Uddevalla, Sweden.

2. Forsberg, K. and L. Dannstedt (2000) "Extensible use of RDF in a business context," *Computer Networks*, Vol. 33, Issues 1-6, June 2000, pp. 347-364. Presented at the *9th International World Wide Web Conference*, Amsterdam, Netherlands.

3. Fagrell, H., K. Forsberg, E. Johannesson and E. Wollerfjord (1999) "NewsMate: Expertise Management for Journalists," Presented at the workshop *Beyond Knowledge Management: Managing Expertise* at the 6th European Conference on Computer Supported Cooperative Work, Copenhagen, Denmark

4. Forsberg, K. and H. Fagrell (1999) "Let's talk about News," In *Proceedings of the 22nd Information Systems Research in Scandinavia (IRIS 22)*, Vol. I, pp. 341-350, Keuruu, Finland.

5. Forsberg, K. and Ljungberg, F. (1998) "The Organising of Editorial Work: Eliciting Implications for New IT use," In *Proceedings of the 21st Information Systems Research in Scandinavia (IRIS 21)*, Vol. 1, pp. 223-236, Sæby, Denmark.

[1] Originally accepted for presentation at the TAGA (Technical Association of the Graphic Arts) 2001 conference in Stockholm, The future of Information and Printed Technologies, October 2001. However, the conference was cancelled.

2. News and newsmaking

The notion of news is ambiguous. Dijk (1988) identifies three different ways 'news' is being used in everyday conversations:

- We talk about new information about events, things or persons ("What is the latest news from your son?").
- We refer to a type of media in which news items are presented ("Did you watch the news?").
- We denote a news story delivered through a news channel, on the web, in the newspaper or on TV ("The story reporting the world record in the World Swimming Championships.")

The concept of news as it has been explored in the present thesis is the last one – news as stories. Bell (1991) pinpoints: "Journalists do not write articles. They write stories. A story has structure, direction, point and viewpoint. An article may lack these." (p. 147)

Traditionally, news stories have been packaged and delivered as news products such as the daily newspaper. The last couple of years, new digital news services have emerged from the conventional news products. News channels have moved from the kitchen tables and living rooms to our desktops and into our pockets (Antikainen et al., 1999). The consumption of news has also changed from being fairly static and often geographically limited to more dynamic and diverse (Eriksen et al., 2000).

Several studies report on how these news services have changed not only the way news stories are presented, but also challenged the way news stories are made:

- The explosion in interactive media forms and changing online journalists roles (Singer, 1998).
- The different production and publishing rhythms (Sabelström-Möller, 2001).
- The customisations of news content (Turpeinen, 2000).

In our interviews and field studies of newsmaking in the MobiNews project we have captured the work practice of journalists in the newsroom and out in the field. The answers we received and the

observations we did made it clear that journalists often have to report events on topics that they do not know very well, with little time for preparation and that reporting is often conducted in the field away from the stationary IT support.

We also found that the newsmaking process, transforming occurrences and happenings of the everyday world into news products and services, can be described in the following way (see figure 3):

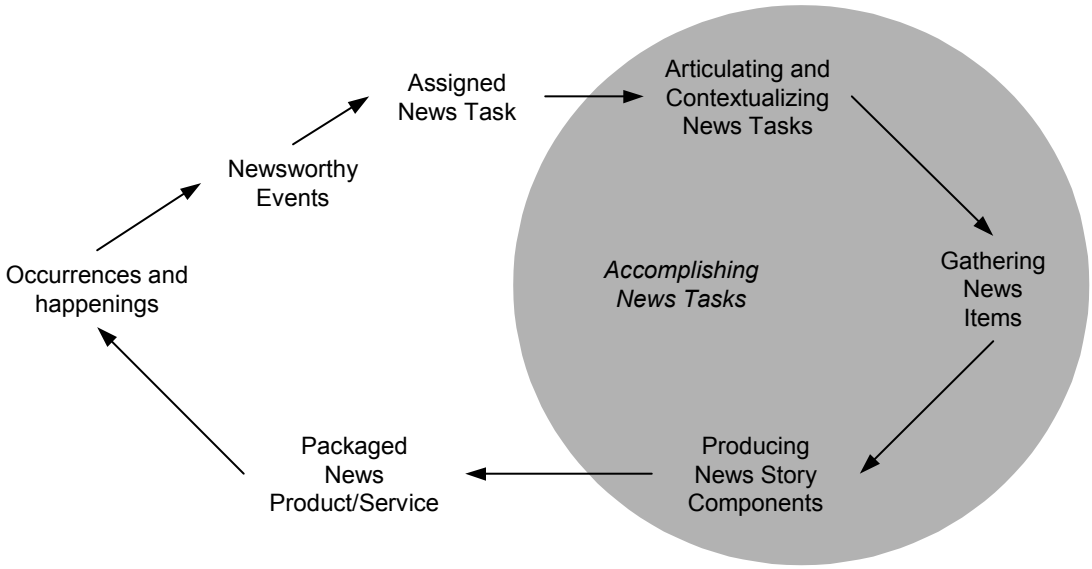


Figure 3: The newsmaking process (see *Paper 4*).

In the broad domain of newsmaking we have focused on the early phase of newsmaking when journalists *articulate news task* (i.e. how they talk about, and more seldom write down, the assigned work) and *contextualise these news tasks* (i.e. how they do background research and frame the news story). This early phase sets the conditions for the gathering of news items, and for the production of news story components. The results are newsworthy stories, covering the selected occurrences or happenings, delivered as packages of content for different media.

In the domain of news research different aspects of newsmaking have been labelled and penetrated in detail by others throughout the years:

- *Gatekeeping*, which is described as a process of reconstructing an event and turning it into news (White, 1950).

- News typification (*hard news* and *soft news*) and *routinization of the unexpected* (Tuchman, 1978).
- *News highlighting* to shape the story content (Gans, 1979).
- *Generating newsworthiness* (Lester, 1980).
- Organisation of time with key events such as the deliverance meeting and *scenic coordination* through cultural means (Kärreman, 1996).
- Local and *global production management systems in newspaper production* and news *product structure* (Nordqvist, 1996).

Newsmaking is definitely cooperative and newsrooms are highly computerised work settings. Nevertheless, Bellotti and Rogers (1997) found a surprising lack of CSCW (Computer Supported Cooperative Work) research on how newsmaking actually gets done and how technology is used in the news and publishing industry. Partly they fill the gap when they describe the cooperative work in newsrooms using the following characteristics: the fast pace of work, the diversity of work, continuous switching between different kinds of representations, the complexity of the collaborative editorial process, constant mobility of workers and the routinisation of work. In this demanding environment they identified the growing diversity of representations, such as dummies of pages, yesterday's newspaper, schedules, news budgets and news assignment lists, for planning, creating and reviewing content (Bellotti and Rogers, 1997).

In recent years, two other CSCW projects have studied work in the news domain, i.e. a study of a Danish radio station (Kensing et al., 1997) and a study of a Finnish newspaper (Helle, 2000).

Kensing et al. (1997) describes the ethnographic and participatory design approaches used in a design project resulting in a computerised Event Calendar and a Program Manager. These two IT services enable a dynamic programming planning process in two dimensions vertical between editorial units and the editorial board and horizontal among the editorial units. They also describe how editorial units, and individual reporters, were reluctant to share information about events to be covered. This was solved through a "make public-

button" to be able to keep a news event or news task private, or local available within the editorial unit.

Helle (2000) presents a systematic analysis of the disturbances in the work practices looking at journalism as part of an activity system including negotiations between different actors, owners, managers, journalists, advertisers etc. Helle (2000) also states "developing computer based support for the integration of tasks needed to produce the daily newspaper could be the next challenge for computer systems design."

None of these studies focus on how journalists continuously access information resources and colleagues to identify, assess, validate, frame and research news tasks on which they are working. Furthermore, previous design-oriented studies do not concern another central topic in newsmaking: mobility.

3. Mobility and context

In our field studies we have focused on the constant mobility of journalists (Bellotti and Rogers, 1997), i.e. the *local mobility* (Bellotti and Bly, 1996) in the newsroom and the *regional mobility*. Reporters leaving the newsroom to localise news in their hometown or in the region covered by their news organisation, developing relationships with sources and participating in external meeting, exemplifies the regional mobility of journalists.

Recently, the issue of mobility has received much attention in the CSCW literature (cf., Luff and Heath, 1998; Kristoffersen and Ljungberg, 2000; Kakihara and Sørensen, 2001).

Luff and Heath (1998) investigated mobility in three different settings: in a medical consultation, at a construction site and in the London Underground. In the case of medical consultation, they identified *micro-mobility* as "the way in which an artefact can be mobilized and manipulated for various purposes around a relatively circumscribed, or "at hand", domain". In the construction site, they recognised *remote mobility*: "individuals that move around different physical locations who require access to information and colleagues".

The London Underground they considered a remote and *local mobility* case.

Kristoffersen and Ljungberg (2000) present three "modalities" of mobile work; visiting, travelling, and wandering. *Visiting* is working in different places for a coherent but limited period, e.g. a journalist leaving the newsroom meeting people in their workplace to do an interview. *Travelling* is working while travelling in a vehicle, such as a bus or a taxi. *Wandering* is working while being mobile locally, i.e. local, physical mobility of users, e.g. a journalist walking away from the morning meeting to participate in a project meeting planning a series of news stories and then stop by a colleague to discuss the framing of a news story.

Kakihara and Sørensen (2001) argue that in addition to the *spatiality* and *temporality* dimensions of mobility, the *contextuality dimension* of mobility is important. The context "... in which the action occurs is of equal importance in organising human interaction; aspects such as "in what way," "in what particular circumstance," and "towards which actor(s)" the action is performed constitute the critical disposition of interaction just as the aspects "where" and "when" do." (Kakihara and Sørensen, 2001)

Nowadays, context in mobile situations often means *context-aware computing* and *context-sensitive applications* in which the concept of context is used to refer to the physical environment in which computational devices are embedded. The classical example of this type of contextualisation, or rather localisation-based feature, is the cell phone that will always vibrate and never beep in a concert, if the system can know the location of the cell phone and the concert schedule. (Moran and Dourish, 2001). However, as Kakihara and Sørensen (2001) argue the contextuality dimension of mobility includes several other aspects than the physical location.

The contextual dimension in general, without taking mobility into consideration, have been explored by others:

Alavi and Leidner (2001) include the *episodic memory* of individuals and groups referring to context-specific and situated knowledge, as one key component in a general model describing knowledge transformation among individuals in a group.

Ackerman and Mandel (1995) use the term *memory in the small* to describe how to combine collective memory and task support. Based

on this design rationale they have developed a small-scale task-based system supporting a scientific community. The stationary system called ASSIST built upon Answer Garden (Ackerman and Malone, 1990), a system designed to facilitate the collection and disseminations of organisational memory. They argue that providing such a task context is more valuable and understandable to users in their everyday work than general efforts to gather entire memory of organisations.

Dahlbom and Mathiassen (1993, p. 25) state, "We want information to be relevant. The information we use should address matters we are concerned with and preferably help us to understand, make decisions, or act."

Fischer (2001) talks about *contextual information* and proposes the following definition of the term: "Information is relevant to the task at hand if it (1) helps someone to understand a specific problem, and (2) is made available when the need for it arises" (Fischer, 2001).

Throughout my research process, I have focused on the contextuality dimension of mobility and in particular on how to provide mobile news reporters with contextual information.

4. Research process

The research question has been addressed using Informatics as the research approach. In particular, I have been influenced by the way Informatics has been applied and described by Dahlbom and Ljungberg (1999) for new and innovative mobile IT services. Mobile Informatics is an application of what has been called the New Informatics. According to Dahlbom (1997, p. 29) the New Informatics is "… a theory and design oriented study of information technology use, an artificial science with the intertwined complex of people and information technology as its subject matter."

The landscape of theories and methods that Informatics covers take us from the initial phase of ethnographic studies, to the final phase including implementation, evaluation and generalisation of new applications. The creative idea generation phase in the middle opens up for scenarios describing desired situations, prototyping and experimenting with ideas for new and innovative IT services realising

the desired situations. The outputs of research efforts that conduct these three phases are (Ljungberg, 1999):

- *Concepts* describing generalisation of the desired situations.
- *Services* programmed to realise desired situations.
- *Documentation* of services and change projects.

A more general discussion of the Mobile Informatics approach can be found in Ljungberg et al. (1998) and Dahlbom and Ljungberg (1999).

I will now describe how I have used Informatics in the newsmaking domain throughout my research process. A process that can be divided into three stages, described in the following sections:

1. The MobiNews research project.
2. Product development in a commercial environment.
3. Revisiting the qualitative research approach.

4.1 The MobiNews research project

The first stage in the research process was conducted as a research project called MobiNews, part of the Mobile Informatics programme. Our overall task was to design new and innovative IT services for mobile news journalism.

The initial phase of ethnographic studies of everyday work were conducted in two Swedish news organisations, one newspaper and one public service radio station.

The first study was initiated from a consultant assignment applying conventional system and process development methods in order to develop and implement an editorial system. However, we felt that these methods, based on the assumption that *what we say we do is what we do,* were not appropriate. We used process-maps but failed to capture the subtle aspects of work practice and collaboration in the newsroom. Therefore, we increased our traditional toolbox with new methods of analysis. We used observations, transcriptions and sociological analysis methods with a primary focus on social interaction to observe and analyse *what we do*, in order to explain the ordinary and obvious everyday actions. We observed everyday work (Hammersley

and Atkinson, 1993) in the newsroom at one of the largest Swedish newspaper (approximately 50 hours).

In 1998 three members of the MobiNews project conducted an extensive field study at a public service radio station during the Swedish election. Three groups of journalist were selected and shadowed both in the newsroom and in the field during three weeks (approximately 300 hours).

All together 350 hours of fieldwork gave us a substantial amount of material; transcriptions of meetings and discussions, copies of documents, photographs of groups and individuals, screenshots and data files from existing applications. A couple of characteristic news events, evolving into news tasks and further on into news stories were selected. We elaborated on these and described the "desired situation" for each one. These situations gave us a starting point for the creative idea generation phase in which we developed services programmed to visualise desired situations. In the phase of implementation and evaluation we have been testing the ideas, re-designed and developed general concepts and architectures.

Informed by the fieldwork an initial prototype system, the NewsPilot, was constructed (Dahlberg et al., 1999) and a generalised design concept, the NewsSpace, was developed (see *Paper 1*). A more ambitious prototype, the NewsMate, was developed (see *Paper 2*) and evaluated in use at Sveriges Radio, Göteborg (Fagrell, 2001). Based on the results a generalised architecture for mobile knowledge management, the FieldWise (see *Paper 3*), was constructed.

4.2 Product development in a commercial environment

A commercial version of FieldWise, adopted for the news domain, has been developed as an add-on product to be integrated with editorial systems. One year of development and marketing in our start-up company forms the second stage in the research process.

As part of the marketing process we demonstrated and discussed FieldWise with a large number of journalists, developers of editorial systems, newspaper managers and investors at several different types of events:

- Presentations at conferences and exhibitions for managers in news and publishing organisations: Dagsvara 2000 Stockholm, IFRA 2000 Amsterdam, Interactive Newspaper 2001 in Dallas, Nordic IFRA 2001 Stockholm.
- Discussions with gurus in the domain of news and publishing technology resulting in journal articles written by Kerry Northrup, IFRA Centre for Advanced News Operations (Northrup, 2000b), Laurel Brunner, European Editor of Seybold (Brunner, 2000) and Gunnar Fahlström, Pressens Tidning (Fahlström, 2000).
- Meetings with groups of journalists and managers in different news organisations such as Svenska Dagbladet, Dagens Nyheter, Norrköpingstidningar, GöteborgsPosten, Mediearkivet, Sveriges Television and Sveriges Radio.
- Technical discussions with developers of different editorial systems.
- Workshops with an advisory board including four persons with substantial knowledge of news work in radio, TV, web and traditional newspaper.

Often we used focus group technique (Nielsen, 1993), where a group of about five to ten people are brought together to discuss issues and concerns about the features of a system. Data from this work was collected as comments, ideas, and proposals. We analysed and evaluated them. We transformed them into requirements and specifications for enhancements of the user interface of FieldWise, technical improvements of the system architecture and businesslike considerations. There were also issues related to the work practice that FieldWise supports.

4.3 Revisiting the qualitative research approach

We found that two of the identified issues once more challenged our traditional methods of capturing the work practice. Together with representatives from two Swedish news organisations, which had found the FieldWise concept interesting, I therefore initiated a field study to derive design improvements (see *Paper 4*). I wanted to explore:

- The ambiguity of 'task' as a term and the articulation of news tasks.
- Sources and type of information gathered during the contextualisation of news tasks.

The field study and the analysis of the observations formed the basis for the proposed improvements of FieldWise. It focused on how the news items evolve and become articulated and contextualised news tasks.

The previously MobiNews studies have positioned the findings in the domain of CSCW and knowledge management (KM) research. This study complemented previous studies since it also relates the findings to theories in the research domain of journalism.

5. Results

In this section, I outline the four papers, highlight the contributions and describe how they together respond to the research question.

5.1 The papers

The first three papers are results from the first stage in the research process conducted in an academic environment. The fourth paper is a result from the last stage of revisiting the academic environment using qualitative research methods.

5.1.1 Navigating in NewsSpace

In the first paper we elicit implications for new ways of using IT when organising editorial work. The empirical studies of editorial staff in a newspaper and in a radio station are summarised in the paper. These studies were done as part of the MobiNews project. For a detailed report from the study of editorial staff at the radio station, including empirical results and implications for design, see Fagrell and Ljungberg (2000).

We also suggest the *NewsCube*; a conceptual model consisting of the three important editorial dimensions; time, content and context. These dimensions summarise our understanding of important factors influencing the organising of editorial work. The news events, news tasks and also the news stories could be positioned in the coordinate system build up by these three dimensions.

As the title of this paper implies, the use of IT supporting the organising of editorial work should offer natural ways of navigating in *NewsSpace*. The design model, or metaphor, of NewsSpace is based on the dimensions covered by the NewsCube. The paper includes ideas for new IT devices and innovative IT services to be used for navigation in NewsSpace;

- *NewsPilot*, a PDA based application investigating how location awareness can support local mobility in a newsroom.
- *NewsBoard*, a NewsSpace device making an 'electronic white-board' part of the editorial system
- *NewsMate*, a PDA based application for knowledge sharing and expertise location among the journalists in both office and field situations

5.1.2 NewsMate: Providing Timely Information to Mobile and Distributed News Journalists

In the second paper we elaborate on the ideas of an application for knowledge sharing and expertise location among journalists. We do so by suggesting NewsMate, a CSCW system that provides mobile and distributed news journalist with timely information such as previous records from internal and external sources, available expertise and plans.

The paper describes a complete use scenario based on a real situation occurring during our field study. The scenario is complemented by a tour through the user interface. The interface was developed using a Windows CE based PDA. For a detailed description of the design and evaluation of NewsMate, see Fagrell (2001).

5.1.3 FieldWise: a Mobile Management Architecture

The third paper describes a generalised architecture for mobile knowledge management, called FieldWise. FieldWise provides mobile access for five client platforms i.e. Pocket PC, Windows CE 2.11 (Palm-sized and Handheld), EPOC, and PalmOS. The requirements that FieldWise is based on are derived from field studies and experiments with the NewsMate prototype. We also describe how Fieldwise is implemented in the news journalism domain.

The empirically grounded requirements for a mobile knowledge management architecture that FieldWise meets are;

- Support evolving and interdependent tasks
- Overview of persistent records
- Support the location of available expertise
- Filtering based on the task and long-term interests
- Dynamic configuration of mobile service

The kind of typical work organisation that can benefit from a FieldWise implementation is one where:

- People's tasks are time critical and driven by deadlines.
- The result relies on the creativity of autonomous, but interrelated people.
- There is a culture of co-operation and sharing of knowledge amongst people.
- People are mobile and distributed.

The domain of work may be, for example, news journalism, sales and real estate brokering. However, every installation requires integration and adaptation to the local work practice of the organisation.

5.1.4 Localising News: Task Articulation and Contextualisation in Mobile News Journalism

The final paper reports from a complementary study conducted to propose improvement of FieldWise. The field study captures the practice of journalists throughout the accomplishment of their news task, in the newsroom and in the field. The observations and analysis elaborate on the following aspects of newsmaking:

- Observing how editors and reporters jointly articulate news tasks in everyday work.

- Analysing specific instances of news tasks captured in the field study.
- Searching for potential categories of news tasks.
- Looking for specifics regarding mobile news tasks.
- Drawing conclusions of how to improve the articulation of news tasks in order to enhance the information gathering for the task-at-hand.

The empirical results reported in the paper include:

- A description of the newsmaking process focusing on the early phase of articulation and contextualisation of news tasks.
- A news task structure, called Localising News, as an example of a newsworthiness strategy that includes activities to find a local reaction, or impact of, national or international event.

Based on the empirical analysis the following improvements of FieldWise are being proposed in the paper:

- Making the newsworthiness strategy explicit, as a structure to define the news task, could improve the way news tasks are being articulated.
- Using the news task structure to organise sources, search queries and information gathered could improve the way the news tasks are contextualised.
- Organising information in such a way that facts about the particular place, time and people are prioritised could improve the way reporters act in mobile situations.

5.2 Contributions

The outputs of the research efforts can be divided into two different types described earlier in this thesis, i.e. theoretical concepts and practical services. The following contributions comprise my answer to the research question describing *how we can provide reporters with contextual information*:

- *Services*, or practical design results, visualising new and innovative IT use:

  - The design and implementation of the *NewsMate prototype*, the first attempt to visualise how to provide mobile news reporters with a task context.

- *Concepts*, or theoretical design implications and general architectures, that bridge the gap from studies of everyday work practice, to design of new and innovative IT services:
  - A *design model,* or *metaphor,* of navigating in the *NewsSpace* using interconnected IT devices to navigate in the interrelated dimensions of time, content and context, both in stationary and mobile settings.
  - *FieldWise, an architecture for mobile knowledge management.* The aim of FieldWise is to support mobile and distributed people conducting tasks that are time critical and driven by deadlines
  - A description of the *newsmaking process* (see figure 2). The early phase of articulation and contextualisation of news tasks has been studied and described in detail.
  - A *news task structure* called *Localising News*, as an example of a newsworthiness strategy, including activates to find a local reaction, or impact, of a national or international news event. Such a task structure could be used to improve the articulation and contextualisation of news tasks.

- *Improved services*, or practical design improvements, enhancing IT services:
  - Proposed *improvements of FieldWise* adopted for the news domain to enhance the IT support of the early phase of the newsmaking process where journalists articulate their news tasks and gather information to be able to deliver newsworthy stories.

6. Discussion

During the process of initiating and formulating the overall design task, elaborating and answering the research question, and verifying and exploiting the results, I have faced a "network of related and dynamically changing contradictions" (Dahlbom and Mathiassen, 1993 p. 60). The following list forms my network of such dialectic contractions:

1. Local work practice vs. business processes.
2. Novel research vs. commercial interesting.
3. Time frames and nature of competition in an academic environment vs. an industrial environment.

Below, I describe the background for these contradictions as my reflections on the research process in the borderline between academia and industry.

1. Local work practice vs. general business processes

In 1997, I wandered into one of the largest newsrooms in Sweden and became fascinated by the frenetic collaborative work. My task as a consultant was to specify new functionality in the editorial system and to define and to facilitate changes in the editorial process. Our objective was to design IT support *optimising for general defined business processes*.

In the research community of CSCW, as well as among advocates of the research approach of Informatics, it is a common understanding that IT support must be designed *anticipating the local practice of work*. As consultants and practitioners, we decided to empirically ground the requirements by conducting a field study of local work practice.

However, such ethnographic studies (cf., Hammersley and Atkinson, 1993) and qualitative evaluation and research methods (cf., Patton, 1990) are seldom used outside the academic community and therefore it is difficult to get acceptance for applying them in a commercial environment.

2. Novel research results vs. commercial products

During our field study it became natural to consider the mobility dimension of peoples work and interaction. We took questions and issues related to mobility in the knowledge intensive environment of news organisations and elaborated on them in a research context. We initiated the MobiNews project. The objective for the project was to design novel and commercially interesting IT support for knowledge management in mobile work.

From an academic perspective, the results from our research are documented as *novel*, due to our publications in the research communities of CSCW and Informatics. From an industrial perspective, the results were recognised as *commercially interesting*, as people spontaneously came up with new professional areas and different situations where the FieldWise concept could be useful.

However, there are many critical steps required to make novel research prototypes become sellable products. The following two contractions exemplifies this:

- The difference between *general potentials* enabled by design concepts and a generic architecture, and the *particular requirements* needed for domain specific adoptions and implementation of it.
- The chasm between the *innovators and early adopters*, and the *early majority* (see Moore, 1991). The diversity between discussing the *technology potentials and competitive advanta*ges appreciated by innovators and early adopters, and providing the *solid references and safety measures* that the early majority ask for to actually buy high-tech products.

3. Time frames and nature of competition in an academic environment vs. in an industrial environment

During four years in the borderline between academia and industry I become aware of the difficulties to define, and follow, a research plan with a time frame of at least one to two years. While the time frames in consultant assignments and product launch plans are most often expressed in months.

It has also been interesting, and sometimes frustrating, to gradually understand the different driving forces and to become aware of the differences in the natures of competition. In an academic environment contributions to the research community are valued, i.e. being reviewed, published and referred. In a commercial environment characterised by consulting and facilitating, other qualities are important such as customer relations, earned money, timeliness, agility and intellectual flexibility. This could also be contrasted with what are seen as key in a commercial environment characterised by development and production. Here the focus is on the collective effort resulting in products or services with sufficient quality, time-to-market, price/performance etc. (See Börjesson et al. (2000) for our discussion on the topic of different values in academia, industry and consulting firms.)

In this thesis, I have described contributions, both on a theoretical and a practical level. Together they comprise my answer to the research question of *how to provide mobile reporters with contextual information*. In addition, I have reflected upon contractions revealed throughout the research conducted in the borderline between academia and industry.

7. References

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